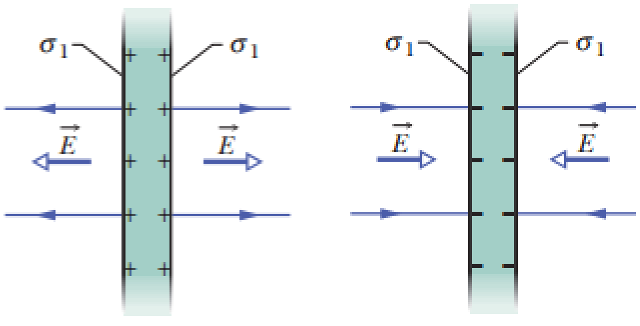
Recap: Electric Field of Two Conducting Sheet

Gauss' Law $\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{enc}}{\varepsilon_0}$

The electric field outside the plate is zero.

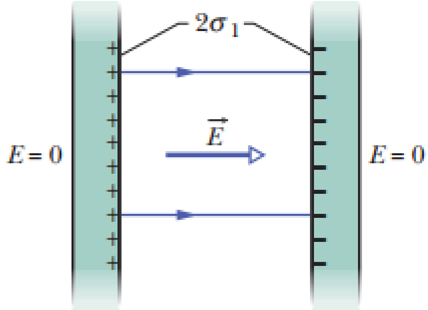


(a) (b)

Between the plates:

$$E = \frac{\sigma_1}{\varepsilon_0} + \frac{\sigma_1}{\varepsilon_0} = \frac{2\sigma_1}{\varepsilon_0}$$

If they are close, charges will be redistributed



(c)

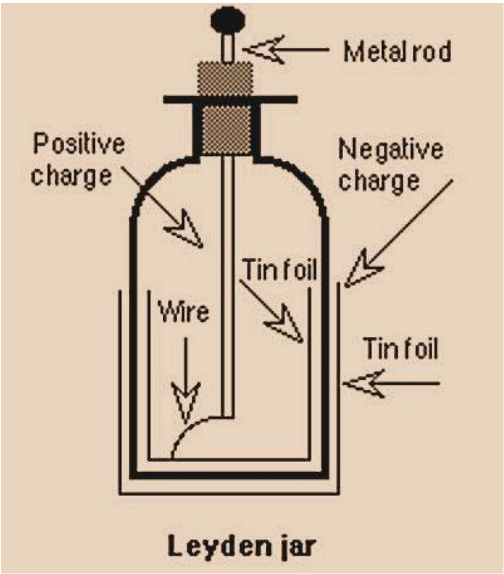
Define $\sigma = 2\sigma_1$:

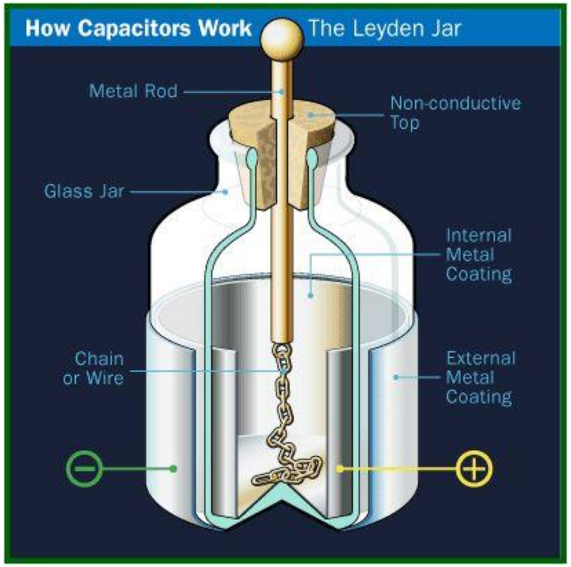
$$E = \frac{2\sigma_1}{\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$$

Electric field in a capacitor

First Capacitor

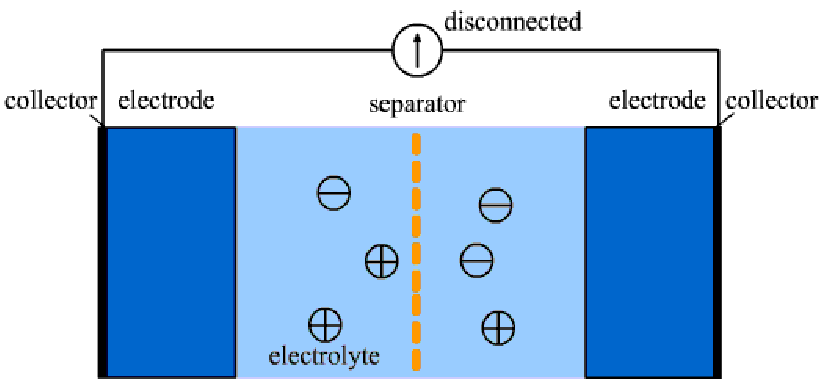
Pieter van Musschenbroek (1692 – 1761) was a Dutch scientist. He held positions in mathematics, philosophy, medicine, and astronomy. He is credited with the invention of the first capacitor in 1746: the **Leyden jar**.



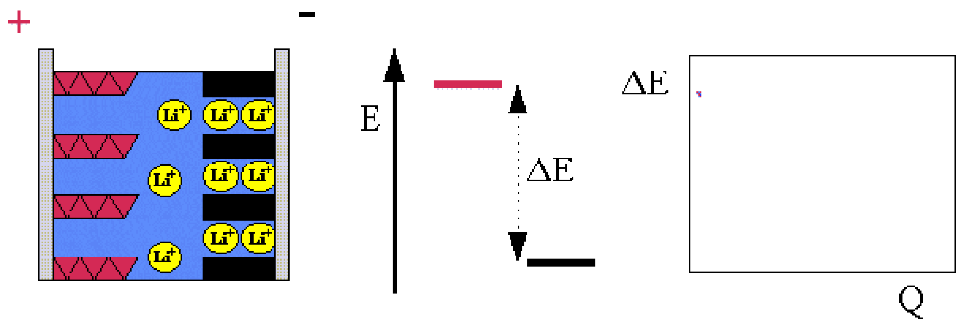


Ch25 Capacitance (电容)

Supercapacitor



Li-battery



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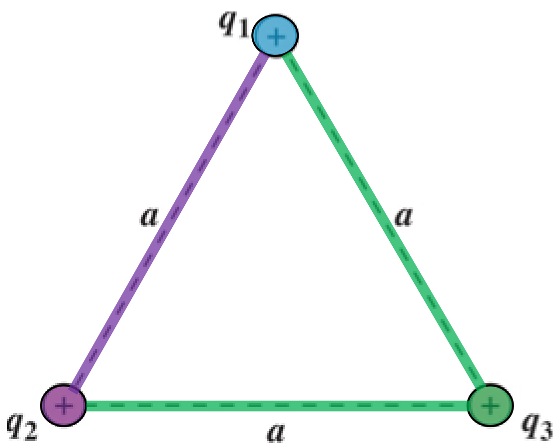
Today's topics (Chapter 25)

- Capacitor and Capacitance
- Energy Stored in Capacitor
- Capacitor with Dielectric Material

Ch25 Capacitance (电容)

Electrostatic Energy = work done to assemble the charge configuration

Reference (0 energy):
when all charges are widely separated.



Bringing q_1 in place takes no work. $W_1 = q_1 V_0 = 0$

Bringing in q_2 takes $W_2 = q_2 V_1(\mathbf{r}_2) = q_2 \, k \dfrac{q_1}{a}$

Bringing in q_3 takes $W_3 = q_3 \left[V_1(\mathbf{r}_3) + V_2(\mathbf{r}_3) \right]$

$$= q_3 \left(k \frac{q_1}{a} + k \frac{q_2}{a} \right)$$

Total electrostatic energy $U = W_1 + W_2 + W_2 \; = \dfrac{k}{a}\left(q_1 q_2 + q_2 q_3 + q_3 q_1 \right)$

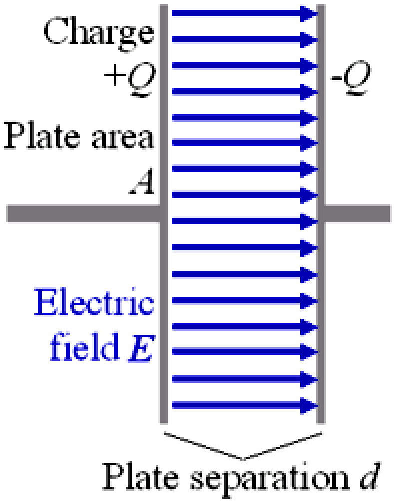
Ch25 Capacitor

- Capacitors are devices that store energy in an electric field.
- Capacitors are used in many every-day applications
  - Heart defibrillators/Camera flash units
- Capacitors are an essential part of electronics.
  - Capacitors can be micro-sized on computer chips or super-sized for high power circuits such as FM radio transmitters.

Ch25 Capacitance (电容)

- Capacitor → Store electrical energy

VS Battery ?



Battery :

- generates a voltage by a chemical reaction, there is no charge separation in a battery until you connect it to something and current flows
- to supply, as far as possible, a constant voltage
- batteries store a lot of charge, but are slow to charge/discharge. Capacitors are the exact opposite: **low storage, fast usage**

Definition of capacitance

The charge and the potential difference V for a capacitor are proportional to each other

$$\boxed{q = CV}$$

The proportional constant C is called the **capacitance** of the capacitor.

Its value depends only on the **geometry** of the plates and not on their charge or potential difference

SI unit: coulomb/volt = **farad, F**

The world's largest capacitor set at Dresden High Magnetic Field Laboratory: ***C* = 0.174 farad**

Why We Define Capacitance in this way?

$$\boxed{q = CV}$$

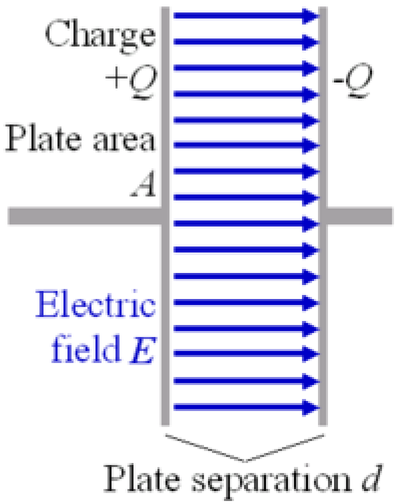
Capacitance describes the capability of storing electric energy. ***q* is straightforward. So why there is *V*?**

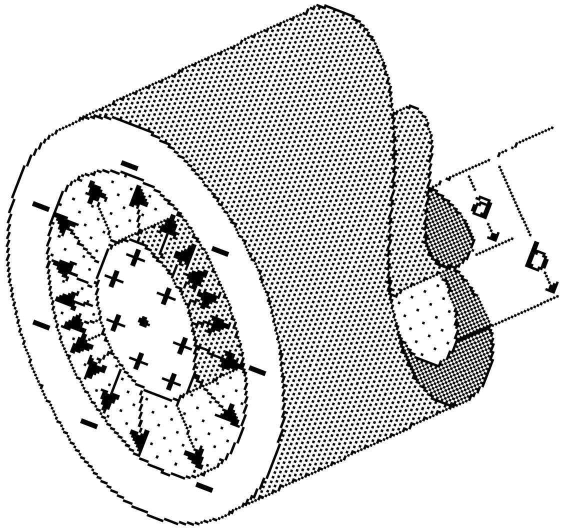
Imagining a high vertical tube to store water. Fill it from the bottom. How much water can you store depends on the pressure you apply to push it in.

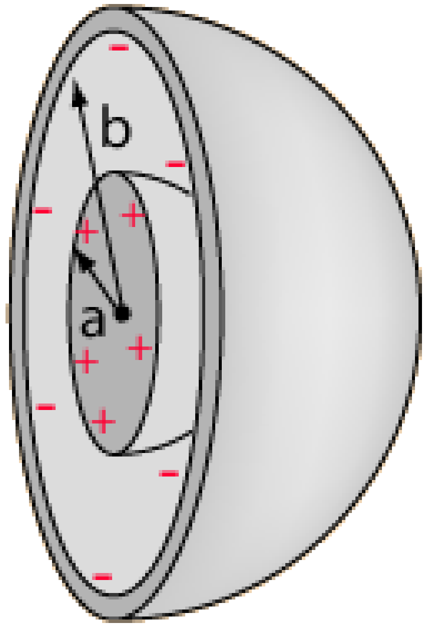
The tube is characterized not the amount of water, but by how easy it is to store the water. That is "capacity".

The capacitor is never full, you can always store more charges, you just have to push harder.

How to calculate capacitance of a given geometry?



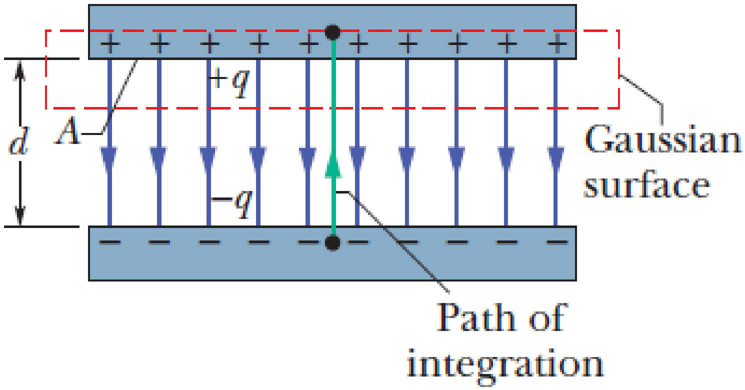




$$\boxed{q = CV}$$

1) Charge on the plate
2) Use Gauss' Law
3) Get $\vec{E}$, get V
4) Get C with $q=CV$

Calculate capacitance of a given geometry



$$\varepsilon_0 \oint \vec{E} \cdot d\vec{A} = q$$

$$q = \varepsilon_0 EA$$

$$V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s}$$

$$V = \int_-^+ E\,ds = E \int_0^d ds = Ed$$

$$\mathrm{C} = \frac{q}{V} = \frac{\varepsilon_0 EA}{Ed} = \frac{\varepsilon_0 A}{d}$$

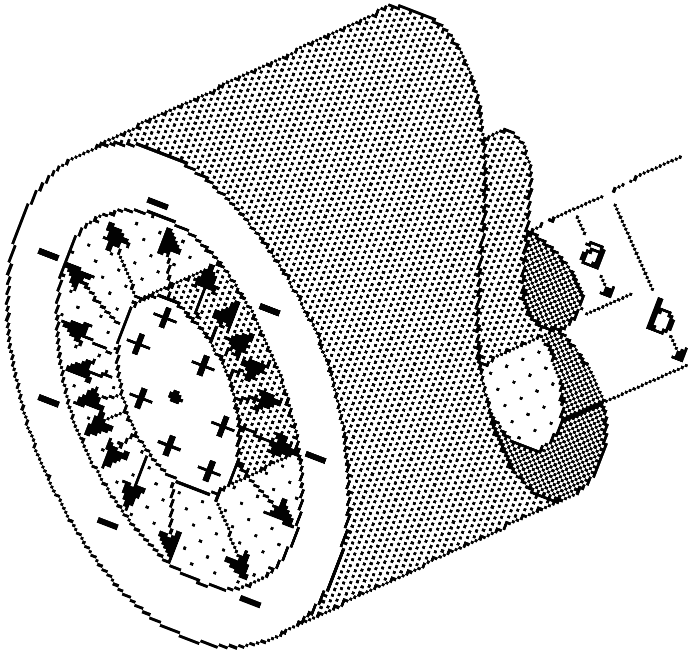
1) Charge on the plate
2) Use Gauss' Law
3) Get ***E*** **,** get ***V***
4) Get C with q=CV

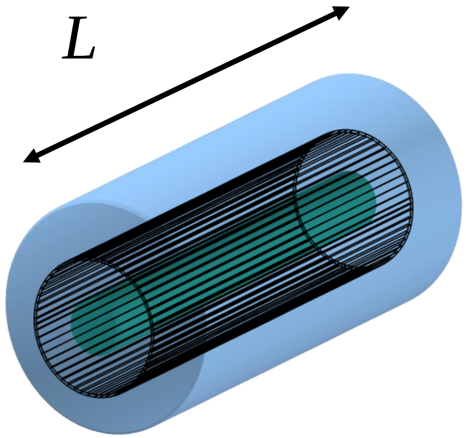
Area $\Uparrow$ ***C*** $\Uparrow$

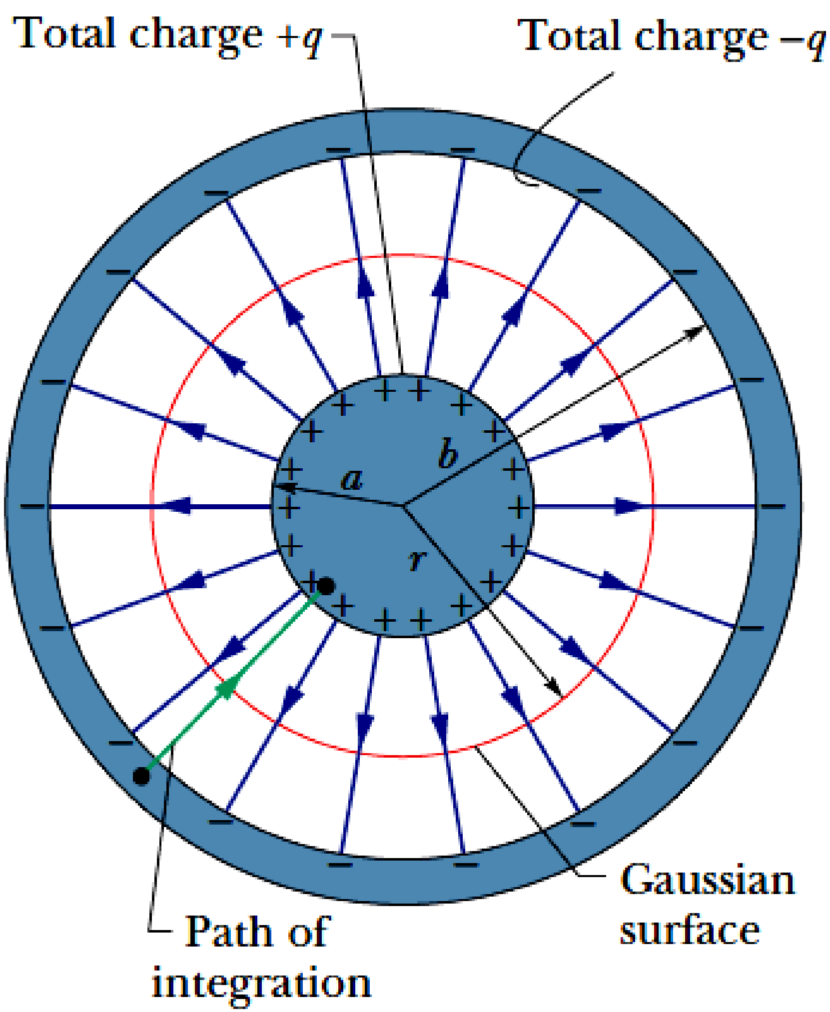
d $\Downarrow$ ***C*** $\Uparrow$

It's all about the geometry!

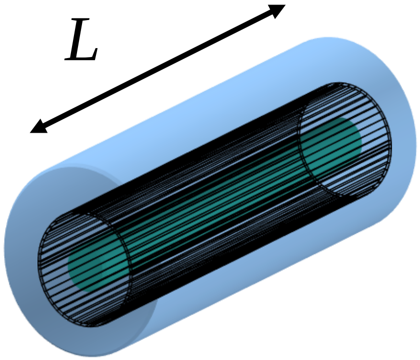
Capacitance of cylindrical capacitor

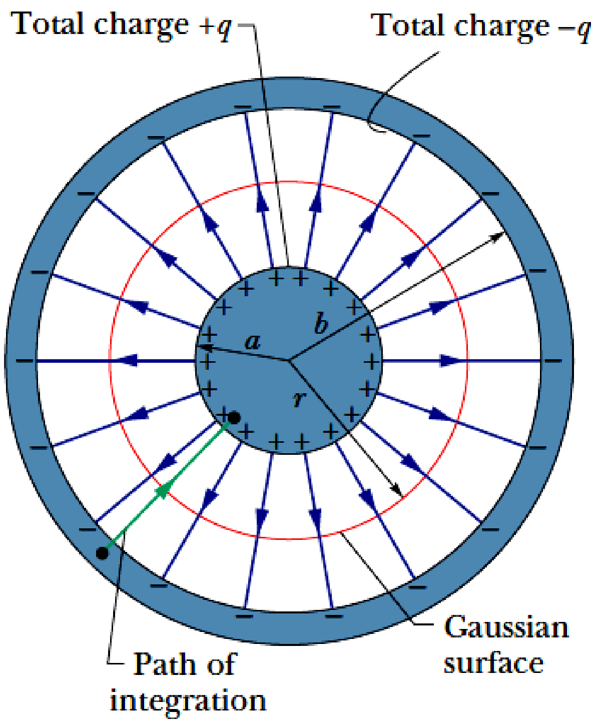






Capacitance of cylindrical capacitor





$$\varepsilon_0 \oint \vec{E} \cdot d\vec{A} = q$$

$$q = \varepsilon_0 EA = \varepsilon_0 E(2\pi r L),$$

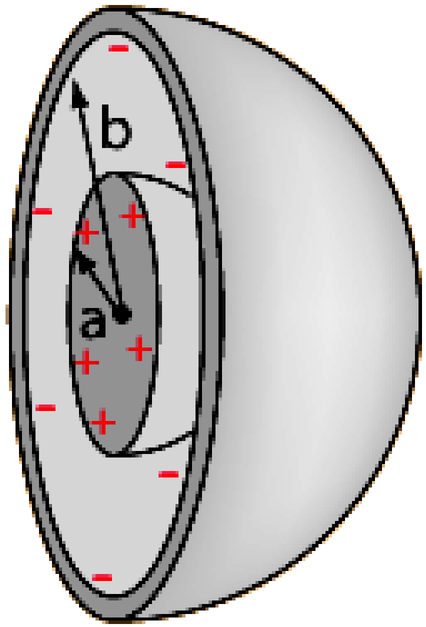
$$E = \frac{q}{2\pi\varepsilon_0 L r}$$

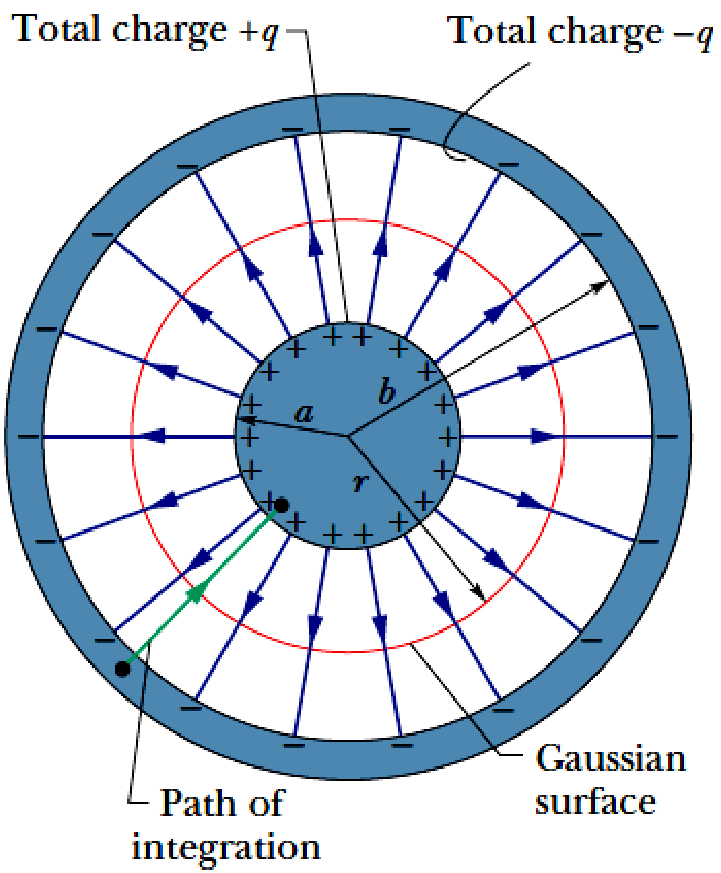
$$V = \int_{-}^{+} E\,ds = -\frac{q}{2\pi\varepsilon_0 L}\int_b^a \frac{dr}{r} = \frac{q}{2\pi\varepsilon_0 L}\ln\left(\frac{b}{a}\right)$$

$$C = q / V$$

$$\boxed{C = 2\pi\varepsilon_0 \frac{L}{\ln(b/a)} \qquad \text{(cylindrical capacitor).}}$$

Calculating the Capacitance, A Spherical Capacitor:





25.3: Calculating the Capacitance, A Spherical Capacitor:

$$q = \varepsilon_0 EA = \varepsilon_0 E(4\pi r^2).$$

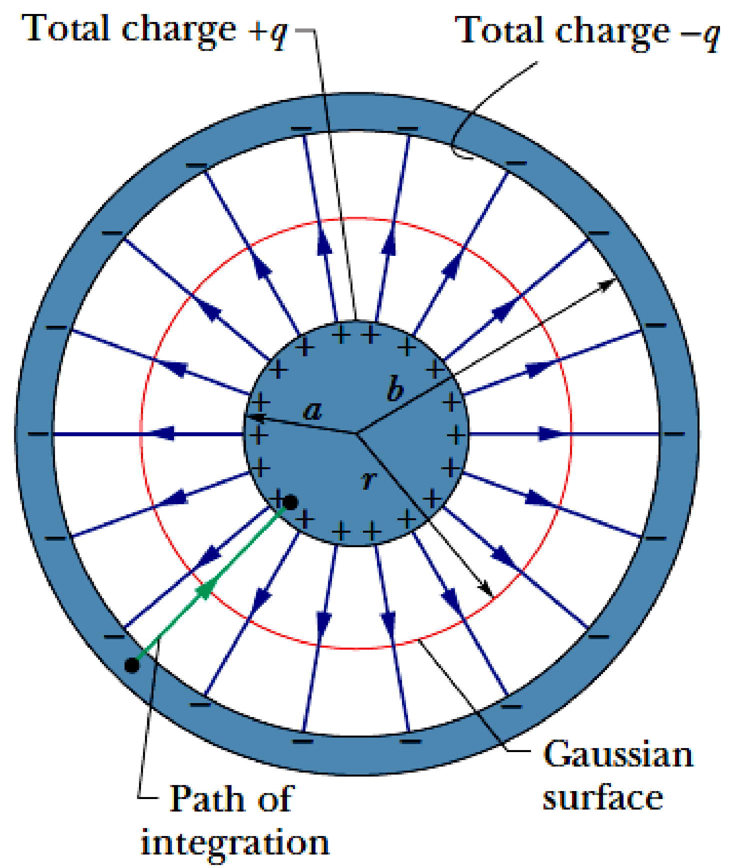
$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$

$$V = \int_-^+ E\,ds = -\frac{q}{4\pi\varepsilon_0}\int_b^a \frac{dr}{r^2} = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{a}-\frac{1}{b}\right)$$

$$= \frac{q}{4\pi\varepsilon_0}\frac{b-a}{ab}$$

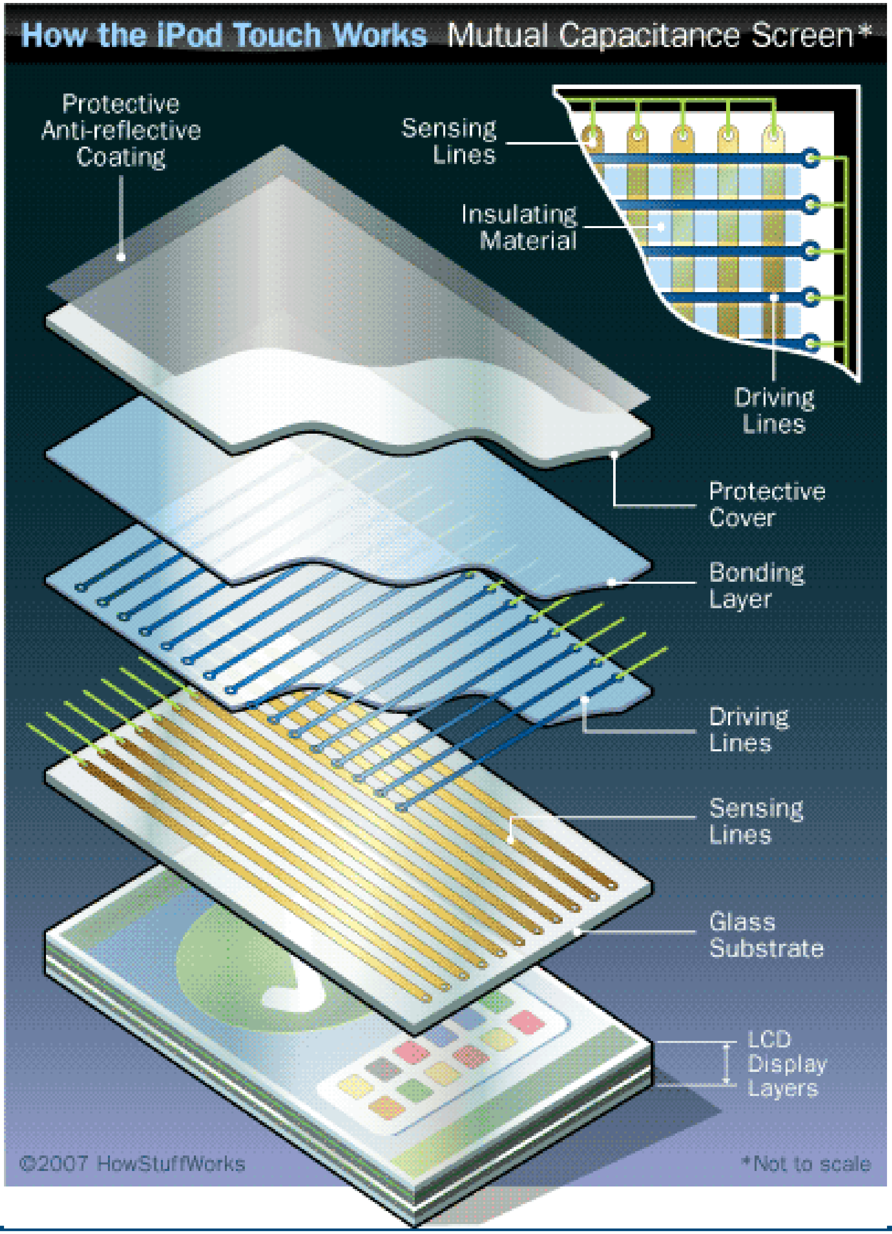
$$C = q / V$$

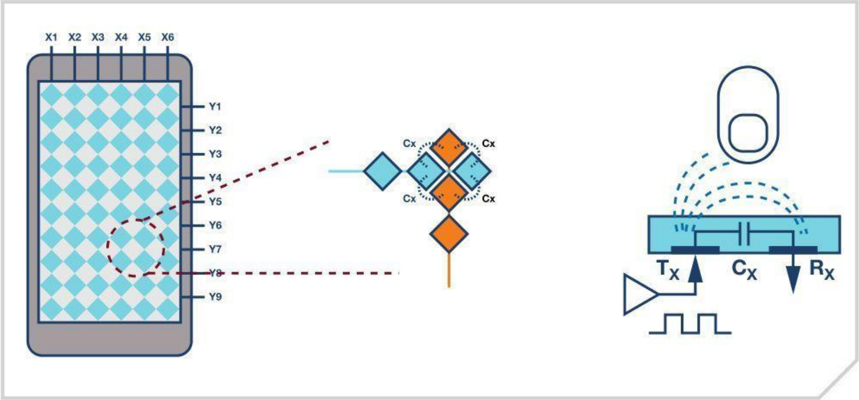
$$\boxed{C = 4\pi\varepsilon_0 \frac{ab}{b-a} \qquad \text{(spherical capacitor).}}$$



What if there is no shell? i.e. only 1 metal sphere

Example: Touch screens of i-Pod Pad Phone





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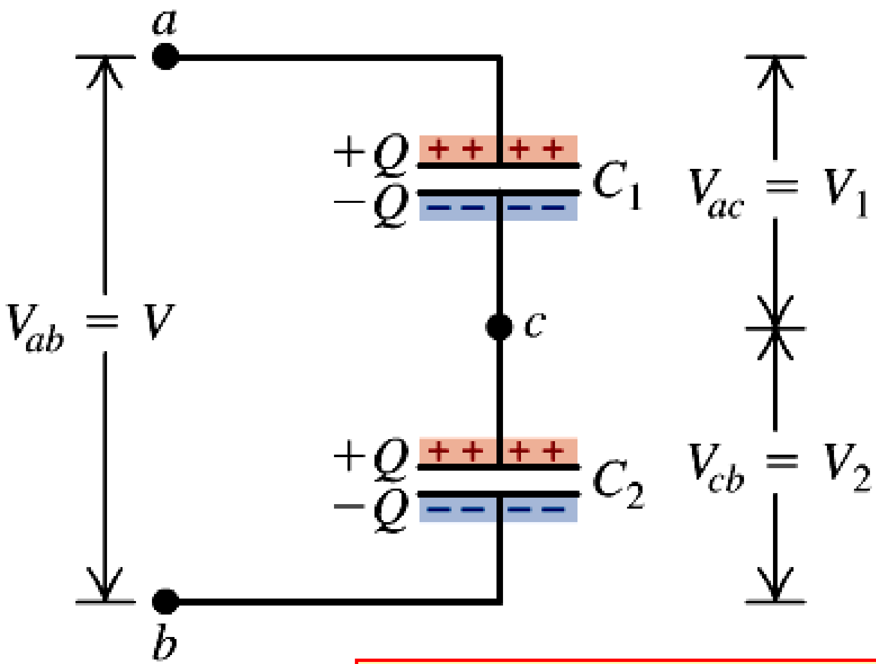
Capacitor in series / parallel connection

$$q = CV$$

Capacitors in series:

- The capacitors have the same charge Q.
- Their potential differences add: $V_{ac} + V_{cb} = V_{ab}$.



$$V_{ac} = V_1 = \frac{Q}{C_1}$$

$$V_{cb} = V_2 = \frac{Q}{C_2}$$

$$V_{ab} = V = V_1 + V_2$$

$$= Q\left(\frac{1}{C_1} + \frac{1}{C_2}\right)$$

$$\boxed{\frac{V}{Q} = \frac{1}{C_1} + \frac{1}{C_2}}$$

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots \qquad \text{(capacitors in series)}$$

Capacitor in series / parallel connection

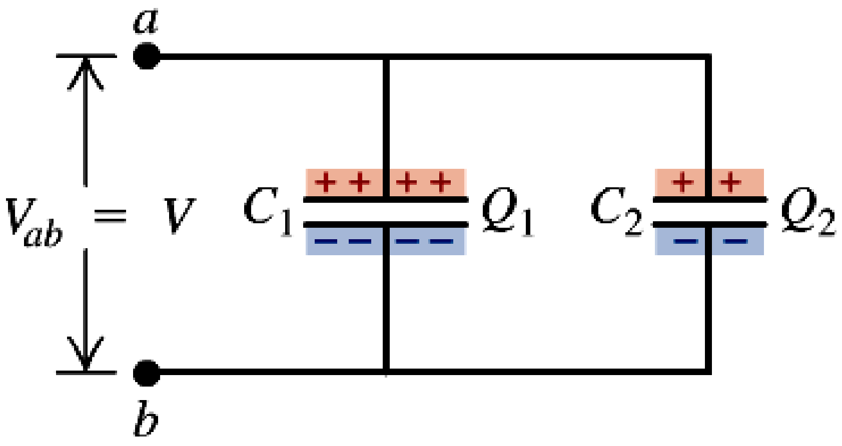
$$\boxed{q = CV}$$

Capacitors in parallel:

- The capacitors have the same potential V.
- The charge on each capacitor depends on its capacitance: $Q_1 = C_1V$, $Q_2 = C_2V$.

$$Q_1 = C_1V$$

$$Q_2 = C_2V$$



$$Q = Q_1 + Q_2$$

$$= (C_1 + C_2)V$$

$$\frac{Q}{V} = C_1 + C_2$$

$$\boxed{C_{eq} = C_1 + C_2 + C_3 + \cdots \qquad \text{(capacitors in parallel)}}$$

Making the Connection

You've got two 10-μF capacitors rated at 15 V.

What are the capacitances & working voltages of their parallel & series combinations?

Parallel :

$$C = 2\times 10\ \mu F = 20\ \mu F$$

$$V_1 = V_2 = V \quad \rightarrow \quad V_{working} = 15\,V$$

Series :

$$C = \frac{1}{2}\times 10\ \mu F = 5\ \mu F$$

$$V_1 = V_2 = \frac{1}{2}V \quad \rightarrow \quad V_{working} = 2\times 15\,V = 30\,V$$

Making the Connection

You have 2 identical capacitors with capacitance C.

How would you connect them to get equivalent

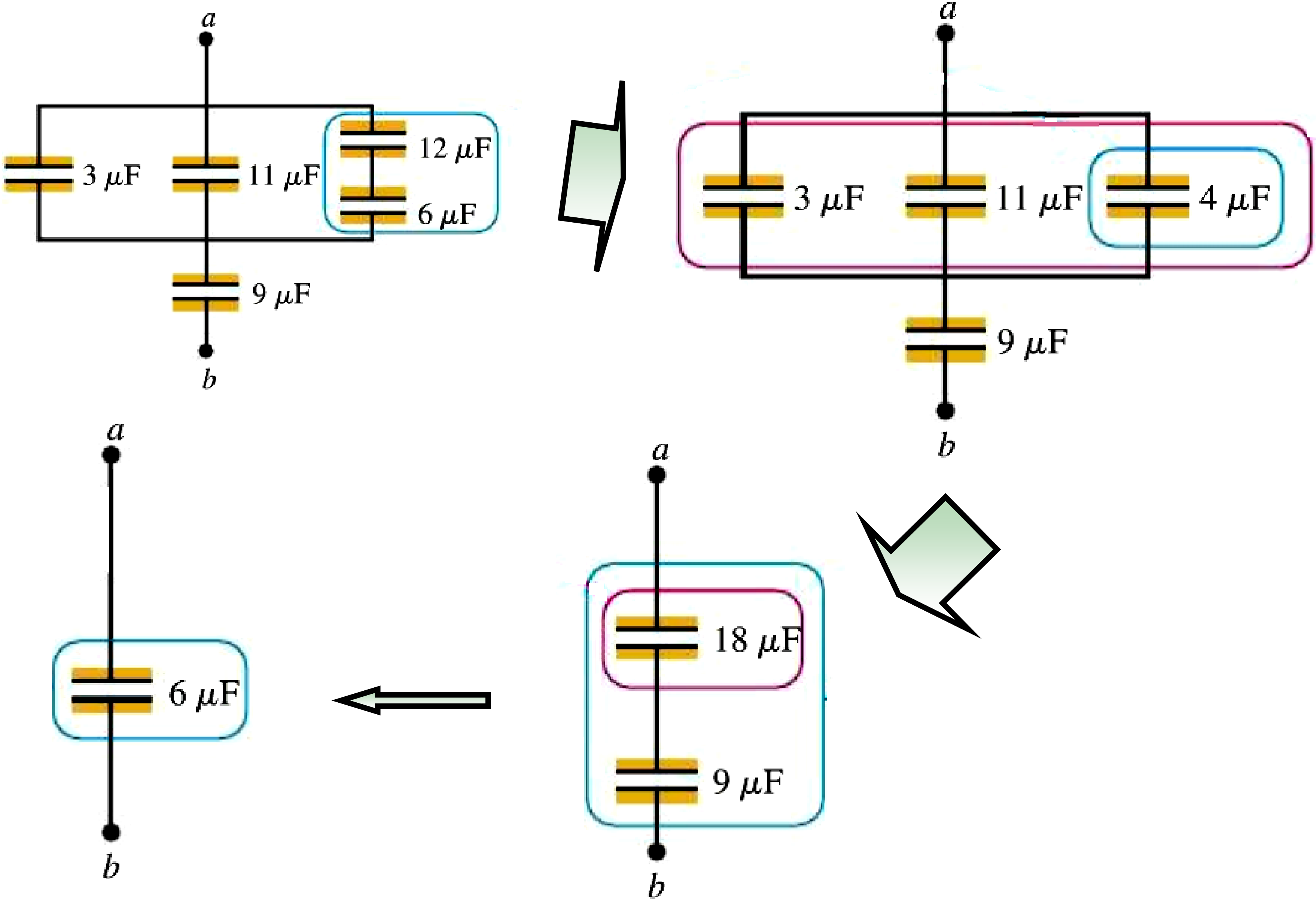
capacitances

parallel (a) 2 C, and

series (b) ½ C ?

Which combination would have the higher working voltage?

Example



Today's topics (Chapter 25)

- Capacitor and Capacitance
- <u>Energy Stored in Capacitor</u>
- Capacitor with Dielectric Material

Energy stored in the capacitor

Energy stored in the capacitor = energy (work done) to charge it

$$dW = v\,dq = \frac{q\,dq}{C} \qquad\qquad v = q/C.$$

$$W = \int_0^W dW$$

$$= \frac{1}{C}\int_0^Q q\,dq$$

$$= \frac{Q^2}{2C} \quad \text{(work to charge a capacitor)}$$

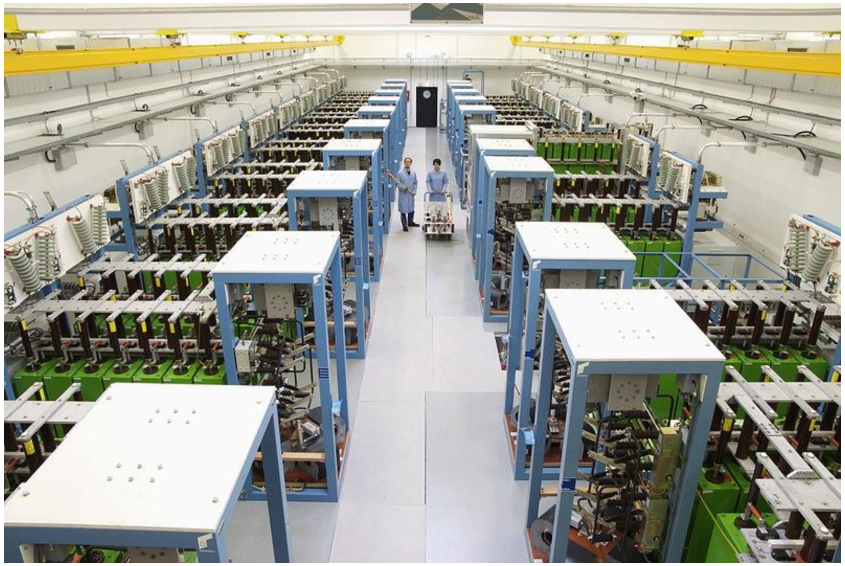
$$\boxed{U = \frac{Q^2}{2C} = \tfrac{1}{2}CV^2 = \tfrac{1}{2}QV}$$

Given a fix ***V***,
a larger *C* means
more energy is stored

World's Largest Capacitor (in Size)

Dresden High Magnetic Field Laboratory (Hochfeld-Magnetlabor Dresden, **HLD**)

A record field close to 94.2 T has been reached in 2011. The aim is to achieve a field of 100 Tesla over a pulse duration of 10 ms. The required energy of 50 MJ is provided by the world's largest capacitor bank, custom-made for this laboratory.



U = 50,000,000 J

V = 24,000 V

$$\Rightarrow C = \frac{2U}{V^2} = 0.173611\ Farads$$

Which Capacitor to use?

A 100-μF capacitor has a working voltage of 20 V,

while a 1.0-μF capacitor is rated at 300 V.

Which can store more charge? More energy?

$$Q_{100\,\mu F} = C\,V \quad = \left(100\times10^{-6}\,F\right)\times\left(20\,V\right) \quad = 2\times10^{-3}C \quad = 2\,mC$$ ☆

$$Q_{1\,\mu F} = \left(1\times10^{-6}\,F\right)\times\left(300\,V\right) \quad = 3\times10^{-4}C \quad = 0.3\,mC$$

$$U_{100\,\mu F} = \frac{1}{2}C\,V^2 \quad = \frac{1}{2}Q\,V \quad = \frac{1}{2}\left(2\,mC\right)\times\left(20\,V\right) \quad = 20\,mJ$$

$$U_{1\,\mu F} = \frac{1}{2}\left(0.3\,mC\right)\times\left(300\,V\right) \quad = 45\,mJ$$ ★

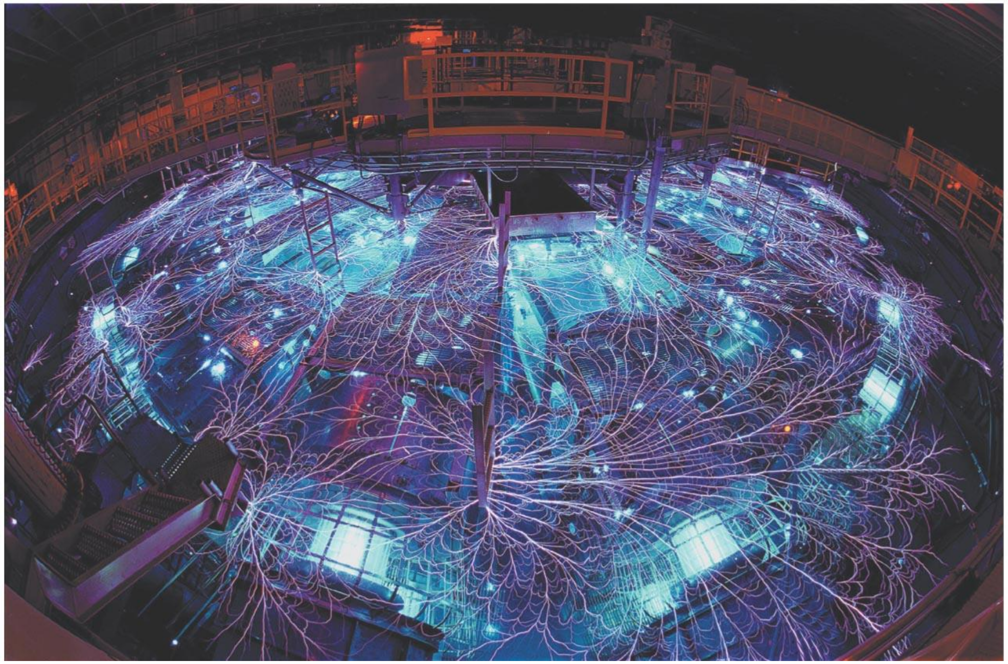
GOT IT?

You need to replace a capacitor with one that can store more energy.Which will give you greater energy increase:

(a) a capacitor with twice the capacitance and same working voltage as the old one, or

☆ (b) a capacitor with the same capacitance and twice the working voltage?

Z machine at Sandia National Lab (New Mexico)

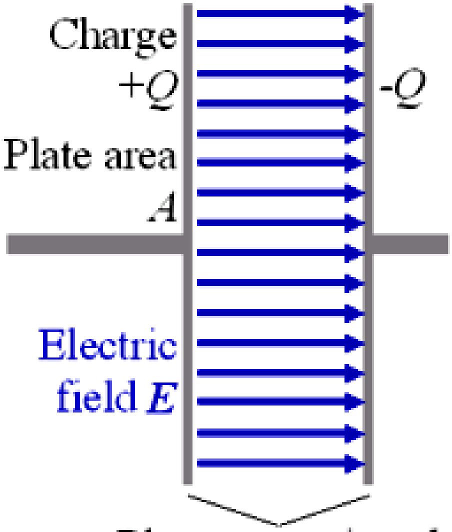


Research for controlled nuclear fusion !

produce up to 2.9×10^{14} W using capacitors in parallel

= 80 times of all the electric power plants on earth!

Electric field energy / Electric energy density



$$U = \frac{Q^2}{2C} = \tfrac{1}{2}CV^2 = \tfrac{1}{2}QV$$

u = Energy density

$$= \frac{\tfrac{1}{2}CV^2}{Ad}$$

$$C = \epsilon_0 A/d$$

$$V = Ed$$

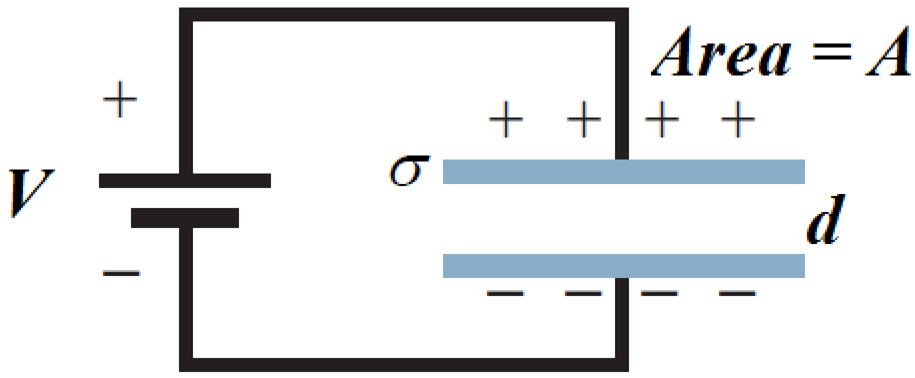
$$u = \tfrac{1}{2}\epsilon_0 E^2$$

(electric energy density in a vacuum)

Energy in empty space! So is it really "empty" ?

Discussion :

The capacitor is kept connected to the battery. If the separation *d* is doubled,



a) How does the electric field change
b) The charge on the plates?
c) The total energy?

$$E = \frac{V}{d}$$

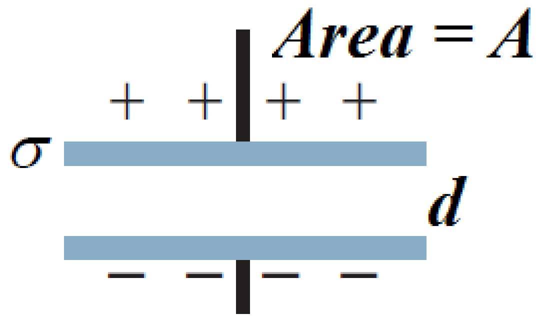
$$U = \frac{q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}\,qV$$

$$C = \frac{\varepsilon_0 A}{d} \qquad \frac{q}{V}$$

Discussion :

The capacitor is disconnected from the battery. If the separation *d* is doubled,

a) How does the electric field change?
b) The potential?
c) The total energy?



$$E = \frac{\sigma}{\varepsilon_0} = \frac{q}{A\varepsilon_0}$$ **E same**

$$E\,d = V$$

V doubles

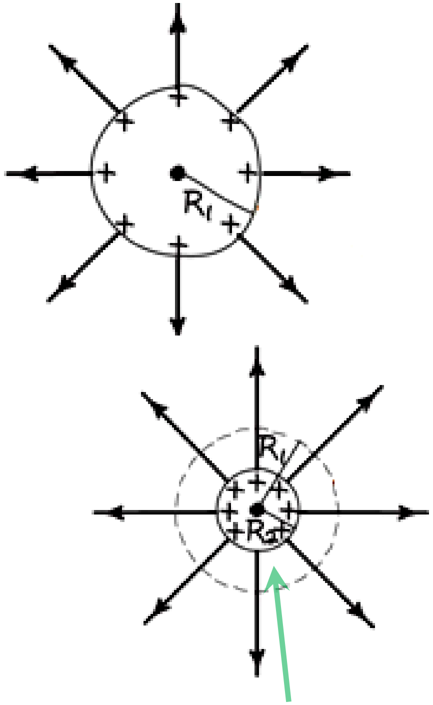
$$U = \frac{q^2}{2C} = \tfrac{1}{2}CV^2 = \tfrac{1}{2}\,qV$$

U doubles

Example: A Shrinking Sphere

A sphere of radius R_1 carries charge Q distributed uniformly over its surface.

How much work does it take to compress the sphere to a smaller radius R_2 ?



Extra energy stored here

$$U = \frac{1}{2}\varepsilon_0 \int \left|\mathbf{E}\right|^2 dV \qquad \mathbf{E} = k\frac{Q}{r^2}\,\hat{\mathbf{r}}$$

$$\Delta U = U_2 - U_1 = \frac{1}{2}\varepsilon_0\left(\int_{R_2}^{\infty} - \int_{R_1}^{\infty}\right)\left|\mathbf{E}\right|^2 4\pi r^2 dr$$

$$= \frac{1}{8\pi k}\int_{R_2}^{R_1}\left(k\frac{Q}{r^2}\right)^2 4\pi r^2 dr = \frac{1}{2}k\,Q^2\int_{R_2}^{R_1}\frac{1}{r^2}dr$$

$$= \frac{1}{2}k\,Q^2\left(\frac{1}{R_2} - \frac{1}{R_1}\right)$$

Work need be done to shrink sphere

$$\Delta U > 0 \quad for \;\; R_2 < R_1$$

Try to solve this problem using the capacitance of a sphere.

GOT IT?

You're at point P a distance $\boldsymbol{a}$ from a point charge $+q$.

You then place a point charge $-q$ a distance $\boldsymbol{a}$ on the opposite side of P as shown.

What happens to

doubles (a) the electric field strength and

quadruples (b) the electric energy density at P ?

decrease (c) Does the total electric energy $U = \int u_E \, dV$ of the entire field increase, decrease, or remain the same?

Negative work is done to bring in –q,
so total electric energy decreases!

More on capacitance: a brief summary

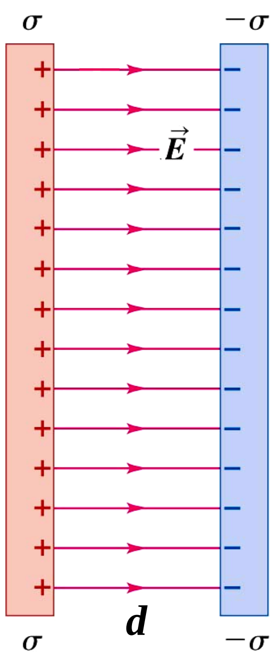
Force $\vec{F} = \dfrac{1}{4\pi\varepsilon_0}\dfrac{qq_0}{r^2}\,\hat{r}$ **Unit : N**

E-field $\vec{E} = \dfrac{\vec{F}}{q_0} = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q}{r^2}\,\hat{r}$ **Unit : N/C**

for uniform E-field $E = \dfrac{V}{d}$ $E = \dfrac{\sigma}{\varepsilon_0}$ **Unit : V/m**

Voltage $V = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q}{r}$ $V = -\displaystyle\int_i^f \vec{E}\cdot d\vec{s}$

$V = \dfrac{U}{q}$ **Unit : V, J/C**



Capacitance $C = \dfrac{q}{V}$ **Unit : F, C/V**

Energy $U = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1 q_2}{r}$ **Unit : J, eV**

for capacitor $U = \dfrac{q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}qV$

Energy density

$u = \frac{1}{2}\varepsilon_0 E^2$

Unit : J/m³

Today's topics (Chapter 25)

- Capacitor and Capacitance
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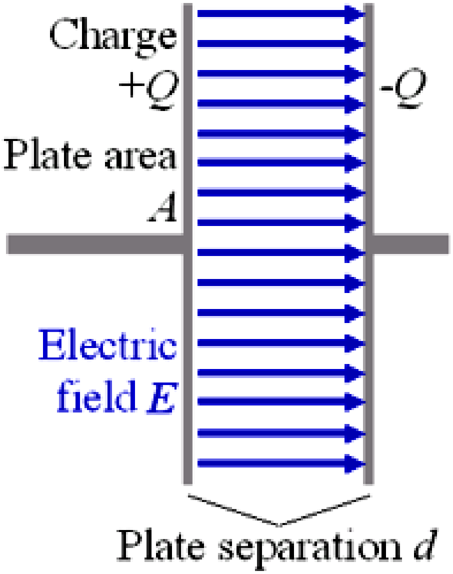
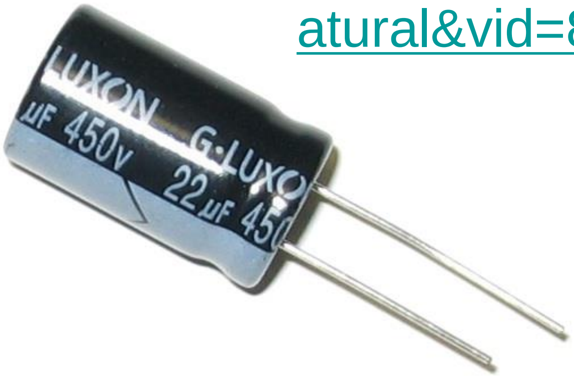
- <u>Capacitor with Dielectric Material</u>

Inside a capacitor





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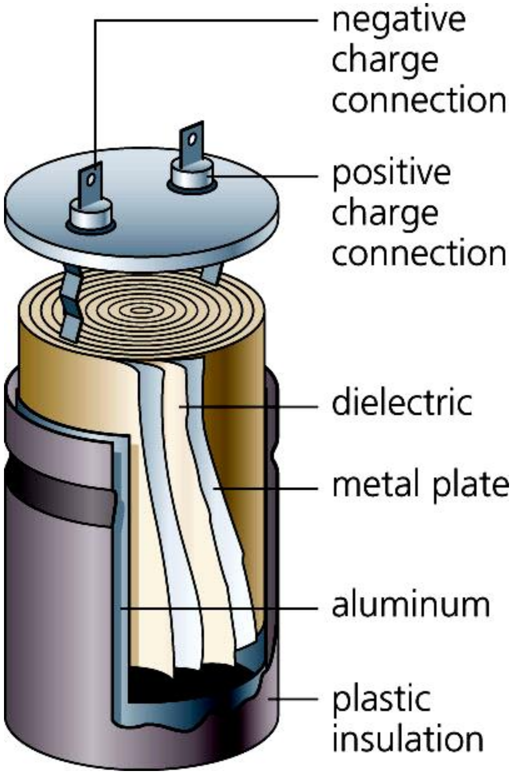
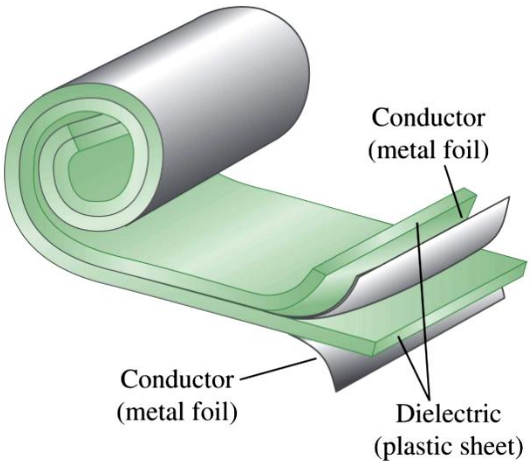


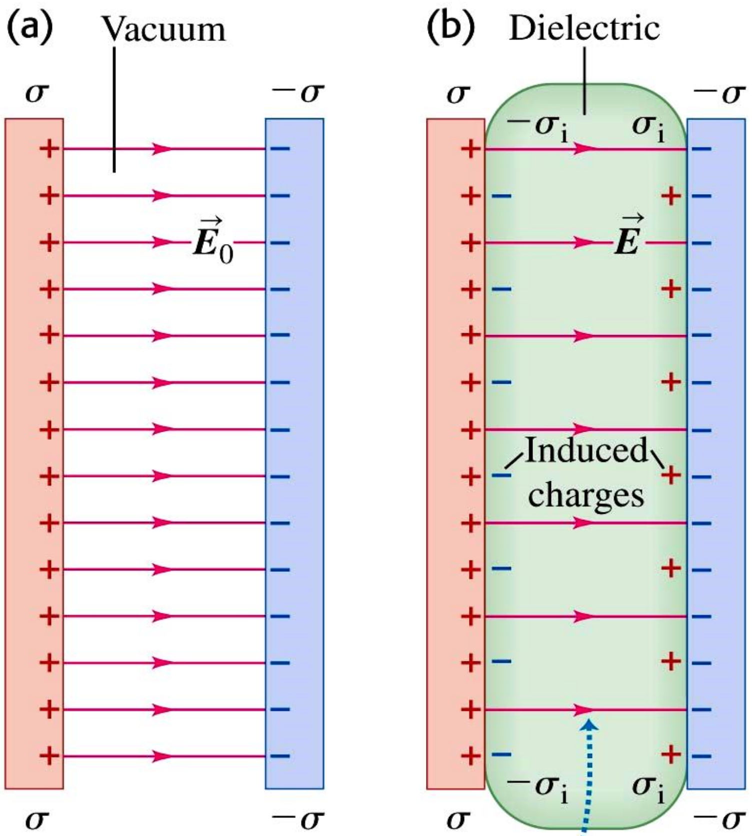
Dielectrics

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{\mathrm{r}}$$

permittivity constant

$\boldsymbol{\varepsilon}_0 =$ 8.85 x10^{-12} $C^2/N\cdot m^2$

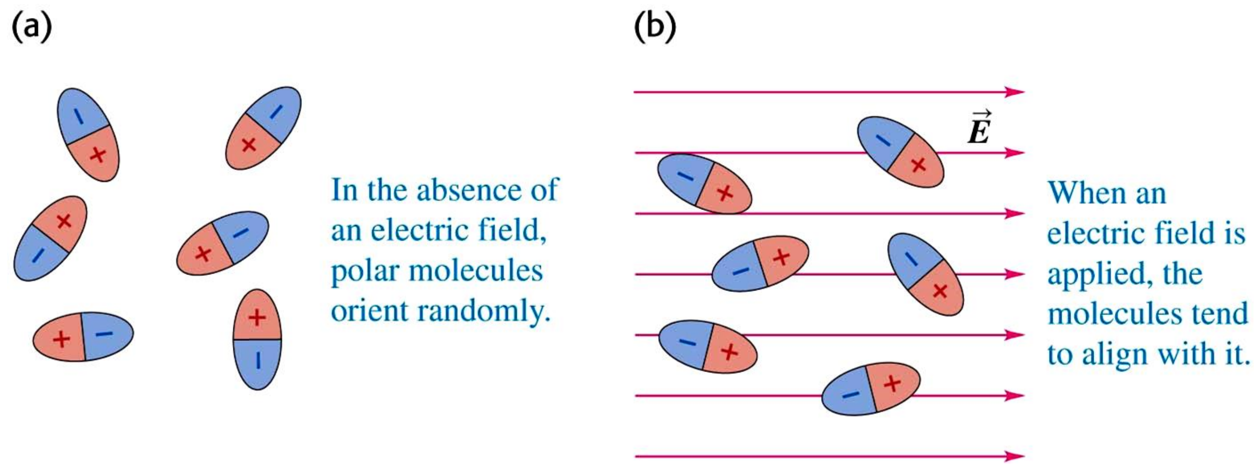




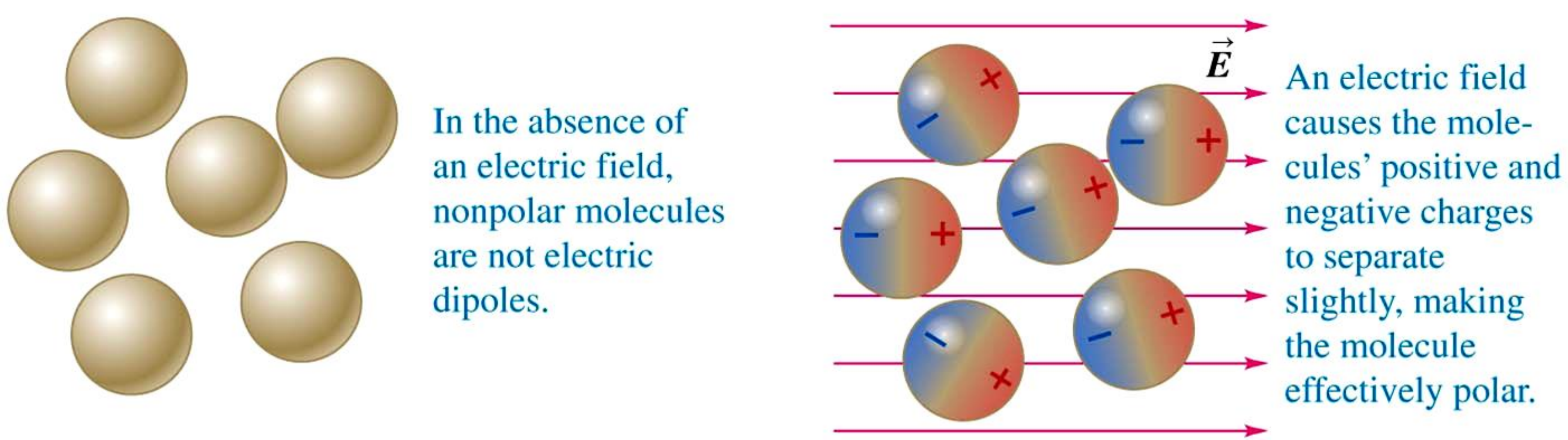
For a given charge density σ, the induced charges on the dielectric's surfaces reduce the electric field between the plates.

Molecular model of induced charge

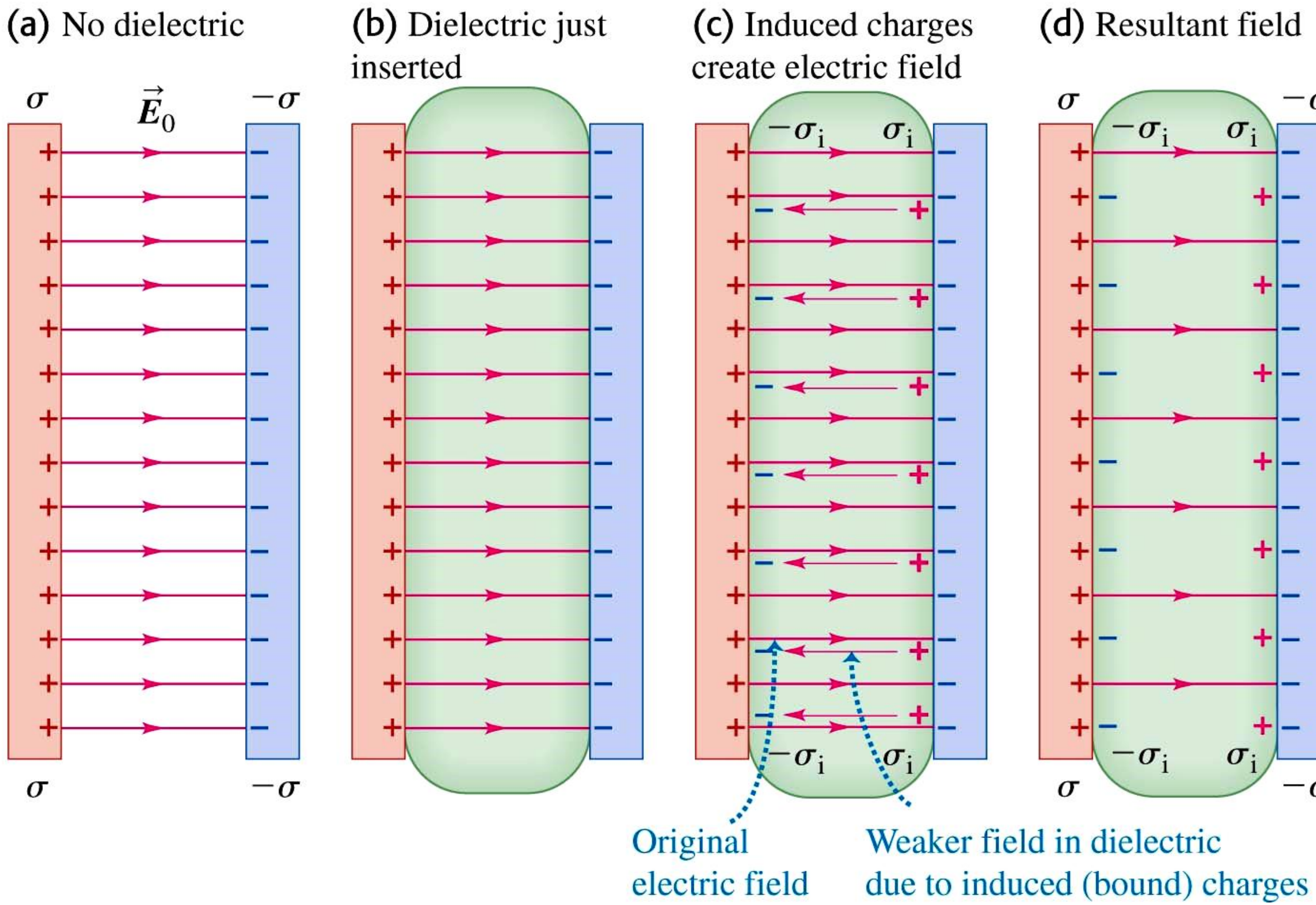
polar molecules



(a) **nonpolar molecules** (b)



Molecular model of induced charge



Dielectrics

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} \Longrightarrow E = \frac{1}{4\pi\kappa\varepsilon_0}\frac{q}{r^2}$$

κ kappa

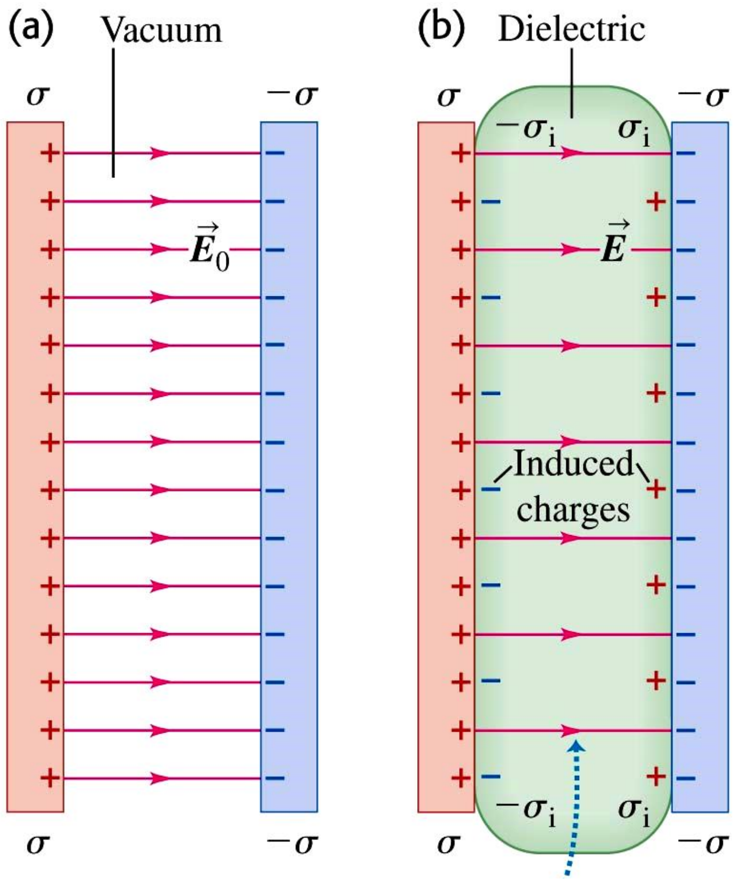
permittivity constant

dielectric constant

Values of Dielectric Constant K at 20°C

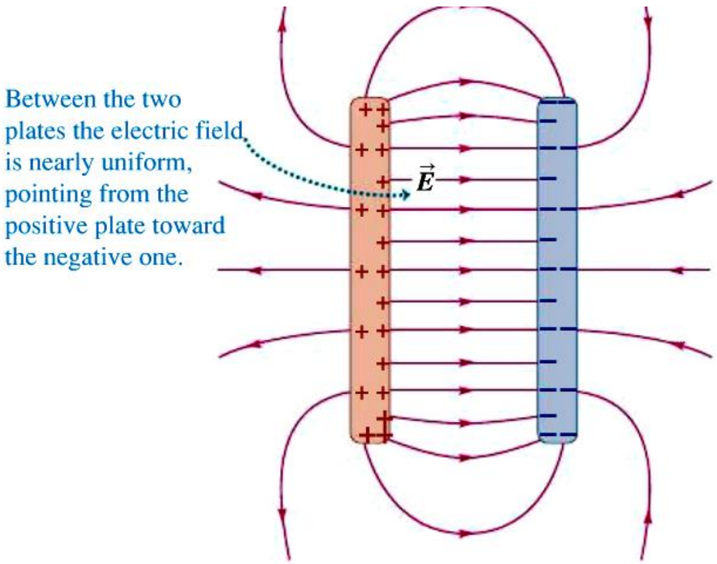
Material	K	Material	K
Vacuum	1	Polyvinyl chloride	3.18
Air (1 atm)	1.00059	Plexiglas	3.40
Air (100 atm)	1.0548	Glass	5–10
Teflon	2.1	Neoprene	6.70
Polyethylene	2.25	Germanium	16
Benzene	2.28	Glycerin	42.5
Mica	3–6	Water	80.4
Mylar	3.1	Strontium titanate	310

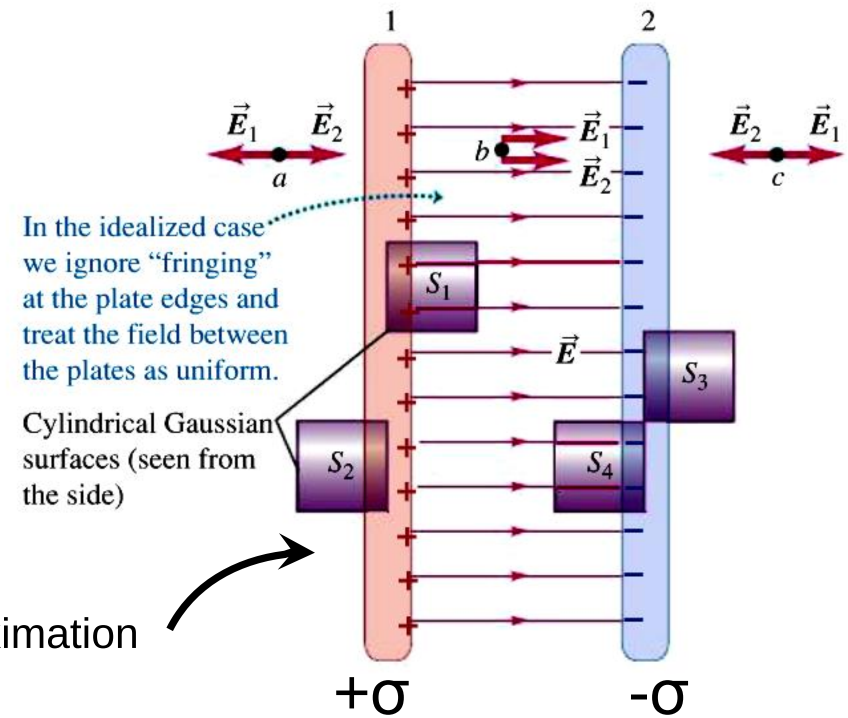
Capacitor with dielectric



How much surface charge is induced in the dielectric?

Case of no dielectric





We will use this approximation

$$EA = \sigma A/\epsilon_0$$

$$E = \frac{\sigma}{\epsilon_0} \text{ (field between oppositely charged conducting plates)}$$

Surface charge density on the dielectric surface

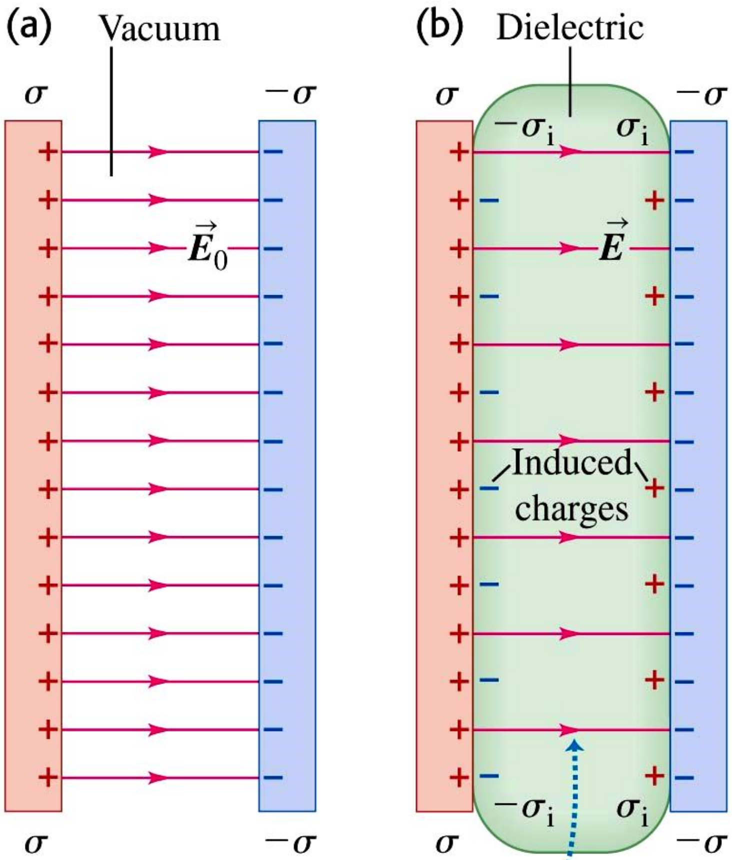
$$E = \frac{\sigma}{\epsilon_0} \text{ (field between oppositely charged conducting plates)}$$

$$E_0 = \frac{\sigma}{\epsilon_0} \qquad E = \frac{\sigma - \sigma_i}{\epsilon_0}$$

$$E = \frac{E_0}{K}$$

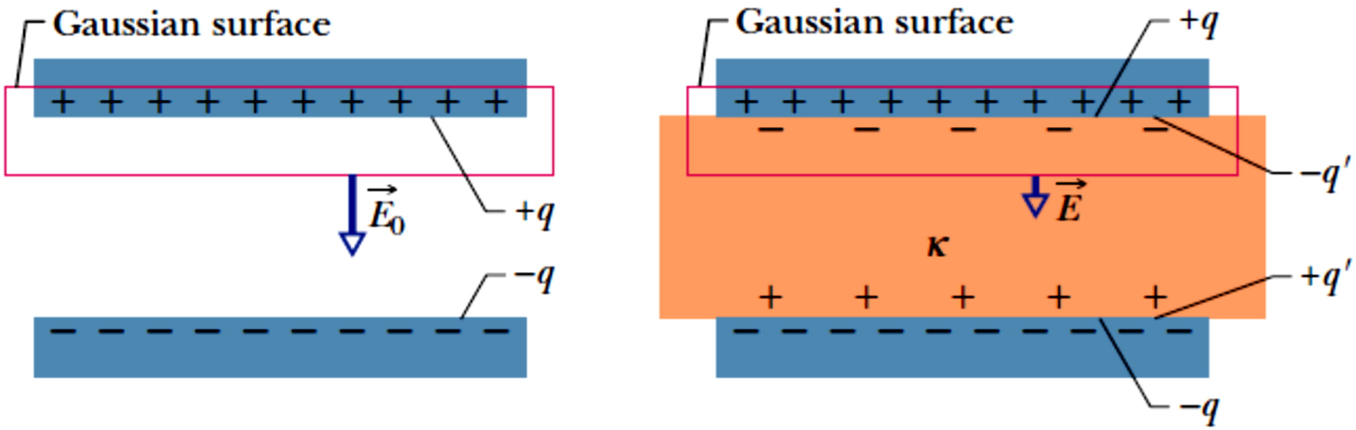
$$\sigma_i = \sigma\left(1 - \frac{1}{K}\right)$$

(induced surface charge density)



For a given charge density σ, the induced charges on the dielectric's surfaces reduce the electric field between the plates.

Dielectric and Gauss' Law



$$\varepsilon_0 \oint \vec{E} \cdot d\vec{A} = \varepsilon_0 EA = q$$

$$E_0 = \frac{q}{\varepsilon_0 A}$$

$$\varepsilon_0 \oint \vec{E} \cdot d\vec{A} = \varepsilon_0 EA = q - q'$$

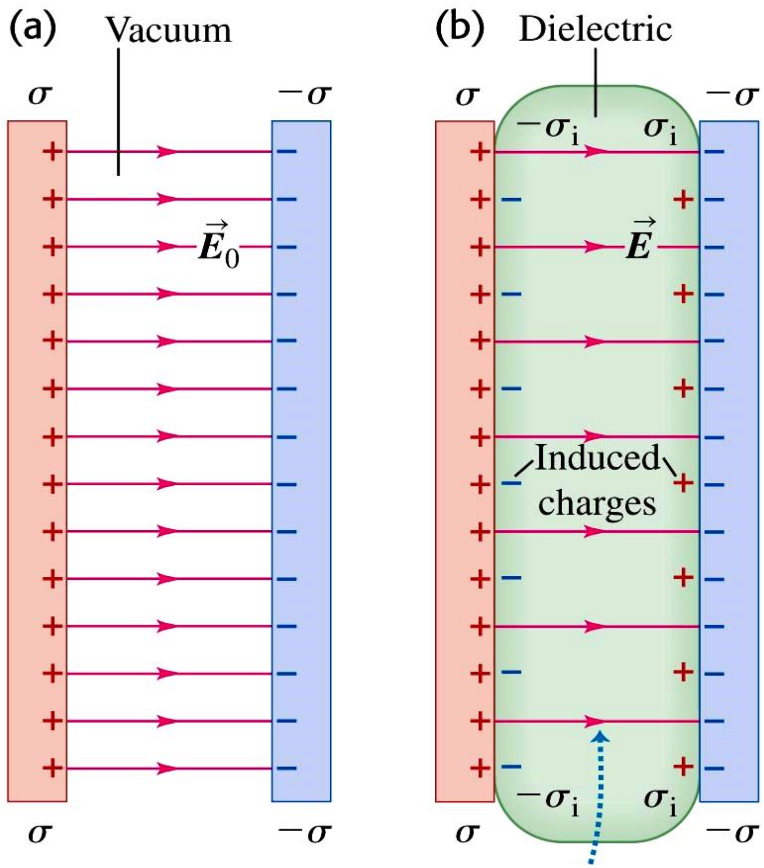
$$E = \frac{q - q'}{\varepsilon_0 A}$$

$$E = \frac{E_0}{\kappa} = \frac{q}{\kappa \varepsilon_0 A}$$

$$\boxed{\varepsilon_0 \oint \kappa \vec{E} \cdot d\vec{A} = q \qquad \text{(Gauss' law with dielectric)}}$$

Dielectrics

Two capacitors storing the same amount of charge Q, but with different dielectrics



$$E = \frac{E_0}{K} \quad \text{(when } Q \text{ is constant)}$$

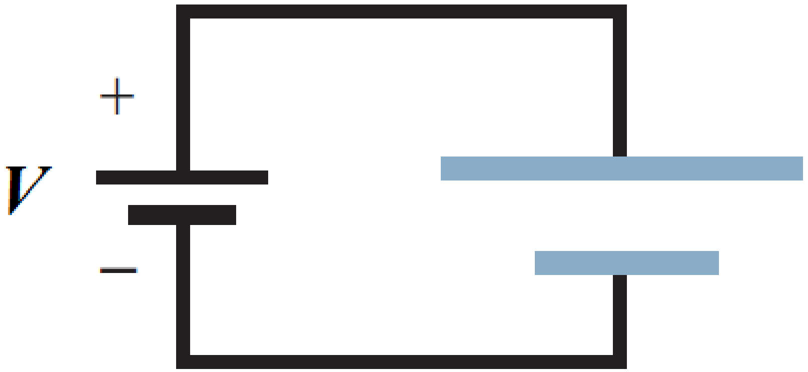
$$V = \frac{V_0}{K} \quad \text{(when } Q \text{ is constant)}$$

$$C = K C_0$$

$$u = \tfrac{1}{2}\epsilon_0 E^2 \Rightarrow u = \tfrac{1}{2}K\epsilon_0 E^2$$

Discussion :

If the upper plate is bigger than the lower plate, do they have the same amount of charge?

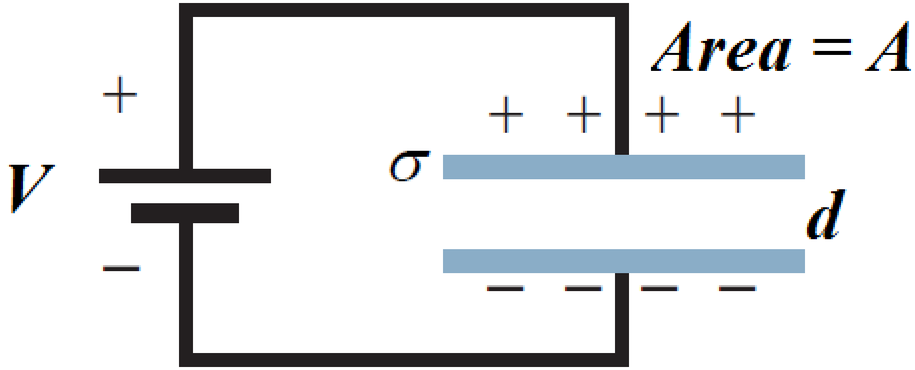


The whole capacitor remains neutral,

So the two plates have charges of equal magnitude

Discussion :

The capacitor is kept connected to the battery. If the separation *d* is doubled,



a) How does the electric field change
b) The charge on the plates?
c) The total energy?

Ans :

$$E = \frac{V}{d}$$

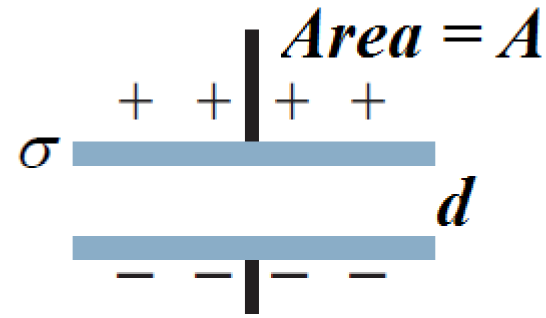
$$U = \frac{q^2}{2C} = \tfrac{1}{2}CV^2 = \tfrac{1}{2}qV$$

$$C = \frac{\varepsilon_0 A}{d} \qquad \frac{q}{V}$$

Discussion :

The capacitor is disconnected from the battery. If the separation *d* is doubled,

a) How does the electric field change?
b) The potential?
c) The total energy?



Ans :

By Gauss' Law, $\varepsilon_0 \oint \vec{E} \cdot d\vec{A} = \varepsilon_0 EA = q$

same *q* **same *E***

$E\,d = V$

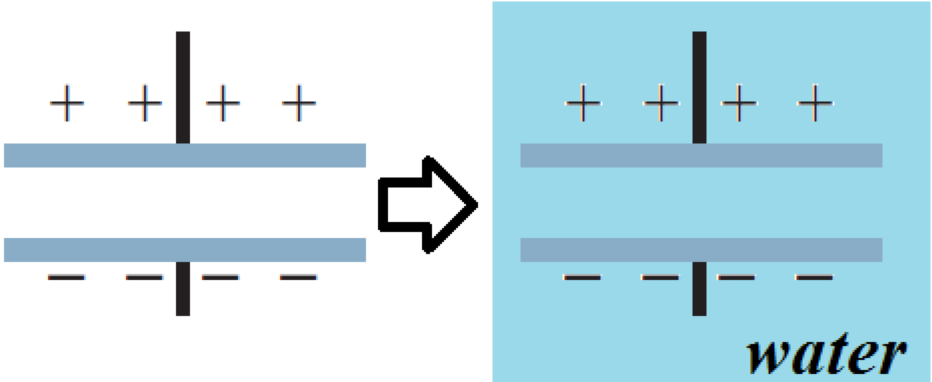
***V* doubles**

$$U = \frac{q^2}{2C} = \tfrac{1}{2}CV^2 = \tfrac{1}{2}\,qV$$

U doubles Why?

Discussion :

When we immerse the charged capacitor into pure water (contains no ion), does the E-field increase, decrease or stay the same?



$$E = \frac{E_0}{K}$$

What if we heat up the water?

Will the dielectric increase or decrease?

Temp. KE of H_2O

which decreases the degree of alignment of their molecular dipoles, thus reducing the E-field they produce to oppose the e-field in the capacitor

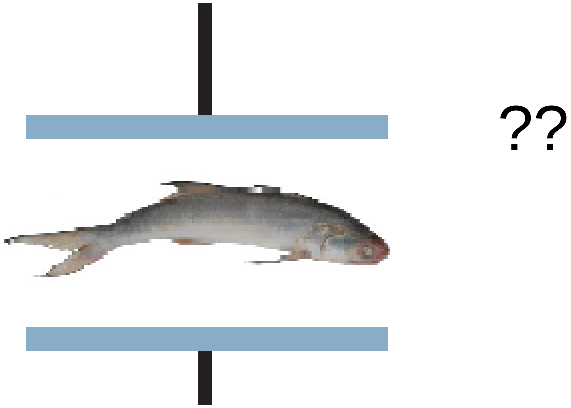
dielectric constant

Discussion

The freshness of fish can be measured by placing a fish between the plates of a capacitor and measuring the capacitance.

How does this work ?

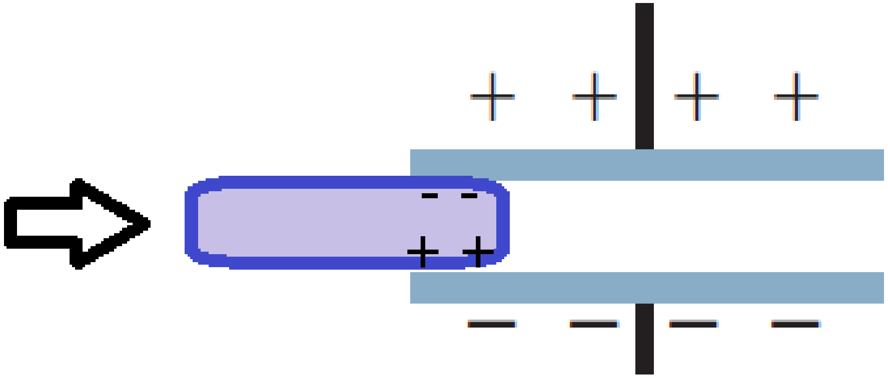
The capacitance depends on the dielectric constant of the fish and this in turn depends on the amount of water in the fish's tissue.



Discussion :

For a charged capacitor, when we bring a slab of dielectric close to the gap between the plates,

a) What force will you feel?
b) What does this force tell you about the energy stored between plates ?



The capacitor exert an attractive force on the dielectric and the potential energy decrease.

$$U = \frac{q^2}{2C}$$

q is constant, C increases ($C = \kappa C_0$)

so U decreases

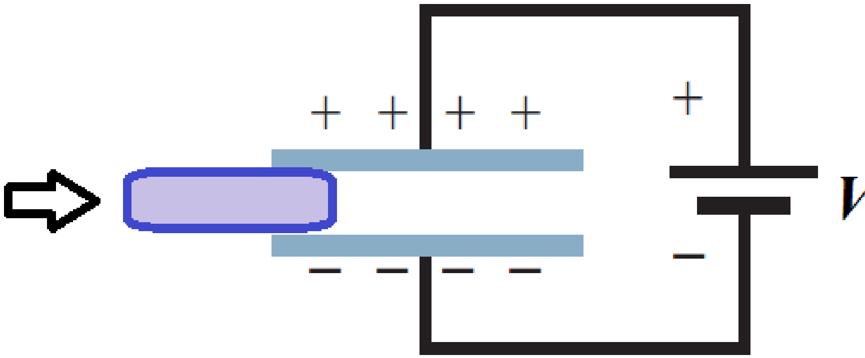
Where does the energy go?

Discussion :

How does the following changes?

a) E-field

E = V / d, V constant, ***E is the same*****.**



b) Charge

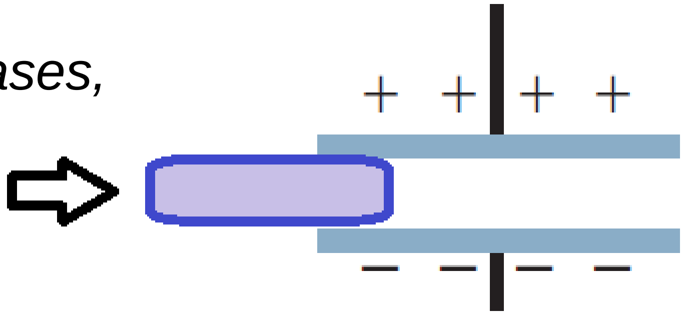
Q = C / V, C increases, ***Q increases*****.**

c) Energy

$U = \frac{1}{2}CV^2$ *V constant, C increases,* ***U increases***

d) E-field

V = Q / C, Q constant, C increases, V decreases,
E = V / d, ∴ E decreases



e) Charge

Q is constant

f) Energy

U = Q^2 / 2C , Q constant, C increases, ∴ ***U decrease***

Classwork

A 20μF capacitor (C_1) is charged to a potential difference of 800 V. The terminals of the charged capacitor are then connected to an uncharged 10μF capacitor (C_2). Find

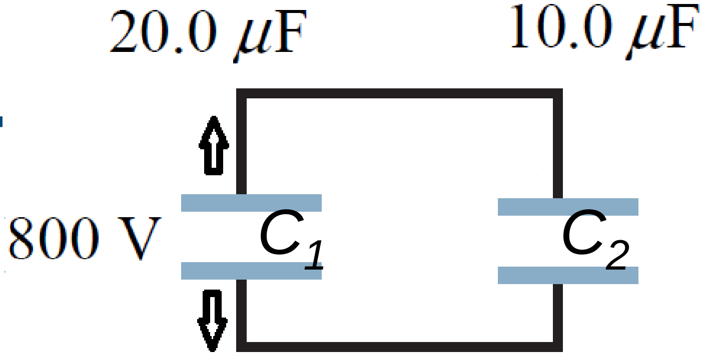
(a) Original charge of the system

(b) Final potential difference across each capacitor

(c) Final energy of the system

(d) The decrease of energy when they are connected

Classwork



a) Original charge of the system

$$Q = C_1(800\ \text{V}) = (20.0\times 10^{-6}\ \text{F})(800\ \text{V})$$
$$= 0.0160\ \text{C}$$

b) Potential difference across each capacitor

$$Q_1 + Q_2 = Q \qquad V_1 = V_2 \qquad V = \frac{Q}{C}$$

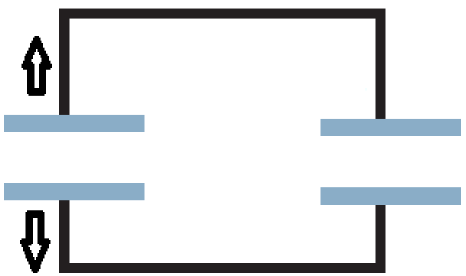
$$\text{gives } \frac{Q_1}{C_1} = \frac{Q_2}{C_2}.\ \ Q_1 = \frac{C_1}{C_2}Q_2 = 2Q_2$$

$$Q_1 + Q_2 = 0.0160\ \text{C gives } 3Q_2 = 0.0160\ \text{C}$$

$$Q_2 = 5.33\times 10^{-3}\ \text{C}.\ \ Q = 2Q_2 = 1.066\times 10^{-2}\ \text{C}$$

$$V_1 = \frac{Q_1}{C_1} = \frac{1.066\times 10^{-2}\ \text{C}}{20.0\times 10^{-6}\ \text{F}} = 533\ \text{V}$$

Classwork



c) Final energy of the system

$$\text{Energy} = \tfrac{1}{2}C_1V^2 + \tfrac{1}{2}C_2V^2 = \tfrac{1}{2}(C_1 + C_2)V^2$$

$$= \tfrac{1}{2}(20.0\times10^{-6}\ \text{F} + 10.0\times10^{-6}\ \text{F})(533\ \text{V})^2 = 4.26\ \text{J}$$

d) The decrease of energy when they are connected

$$\text{initially has energy} = \tfrac{1}{2}C_1V^2 = \tfrac{1}{2}(20.0\times10^{-6}\ \text{F})(800\ \text{V})^2$$

$$= 6.40\ \text{J}$$

$$6.40\ \text{J} - 4.26\ \text{J} = 2.14\ \text{J}$$

conversion of electrical energy to other forms during the motion of the charge

Where does the energy go?

SUMMARY

Review & Summary

Capacitor; Capacitance A **capacitor** consists of two isolated conductors (the *plates*) with charges $+q$ and $-q$. Its **capacitance** C is defined from

$$q = CV, \tag{25-1}$$

where V is the potential difference between the plates.

Determining Capacitance We generally determine the capacitance of a particular capacitor configuration by (1) assuming a charge q to have been placed on the plates, (2) finding the electric field $\vec{E}$ due to this charge, (3) evaluating the potential difference V, and (4) calculating C from Eq. 25-1. Some specific results are the following:

A *parallel-plate capacitor* with flat parallel plates of area A and spacing d has capacitance

$$C = \frac{\varepsilon_0 A}{d}. \tag{25-9}$$

A *cylindrical capacitor* (two long coaxial cylinders) of length L and radii a and b has capacitance

$$C = 2\pi\varepsilon_0 \frac{L}{\ln(b/a)}. \tag{25-14}$$

A *spherical capacitor* with concentric spherical plates of radii a and b has capacitance

$$C = 4\pi\varepsilon_0 \frac{ab}{b-a}. \tag{25-17}$$

An *isolated sphere* of radius R has capacitance

$$C = 4\pi\varepsilon_0 R. \tag{25-18}$$

Capacitors in Parallel and in Series The **equivalent capacitances** C_{eq} of combinations of individual capacitors connected in **parallel** and in **series** can be found from

$$C_{eq} = \sum_{j=1}^{n} C_j \quad (n \text{ capacitors in parallel}) \tag{25-19}$$

and

$$\frac{1}{C_{eq}} = \sum_{j=1}^{n} \frac{1}{C_j} \quad (n \text{ capacitors in series}). \tag{25-20}$$

Equivalent capacitances can be used to calculate the capacitances of more complicated series–parallel combinations.

Potential Energy and Energy Density The **electric potential energy** U of a charged capacitor,

$$U = \frac{q^2}{2C} = \tfrac{1}{2}CV^2, \tag{25-21, 25-22}$$

is equal to the work required to charge the capacitor. This energy can be associated with the capacitor's electric field $\vec{E}$. By extension we can associate stored energy with any electric field. In vacuum, the **energy density** u, or potential energy per unit volume, within an electric field of magnitude E is given by

$$u = \tfrac{1}{2}\varepsilon_0 E^2. \tag{25-25}$$

Capacitance with a Dielectric If the space between the plates of a capacitor is completely filled with a dielectric material, the capacitance C is increased by a factor κ, called the **dielectric constant,** which is characteristic of the material. In a region that is completely filled by a dielectric, all electrostatic equations containing ε_0 must be modified by replacing ε_0 with $\kappa\varepsilon_0$.

The effects of adding a dielectric can be understood physically in terms of the action of an electric field on the permanent or induced electric dipoles in the dielectric slab. The result is the formation of induced charges on the surfaces of the dielectric, which results in a weakening of the field within the dielectric for a given amount of free charge on the plates.

Gauss' Law with a Dielectric When a dielectric is present, Gauss' law may be generalized to

$$\varepsilon_0 \oint \kappa\vec{E}\cdot d\vec{A} = q. \tag{25-36}$$

Here q is the free charge; any induced surface charge is accounted for by including the dielectric constant κ inside the integral.

Brainstorm

Why does the energy stored in a capacitor increase when the dielectric is removed? Where does the extra energy come from?